Swiss Centre
for Life Cycle
Inventories

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ecoinvent data v2.2

the 2010 version of the most comprehensive and most popular public LCI database

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The mission of the ecoinvent Centre is to provide the most relevant, reliable, transparent and accessible LCI data for users all over the world. For this, the ecoinvent Centre provides the database ecoinvent in its version v2.2 with more than 4000 industrial LCI datasets to assist every-day work in areas like IPP, Design for Environment (DfE), Environmental Management Systems (EMS) or Product Stewardship.

Compared to the version v2.1, the following changes have been implemented in this new version v2.2 of ecoinvent:

1. NEW DATASETS:

Almost 150 additional datasets can be found in this latest version of the database ecoinvent. Among them are e.g. …
… new data in the area of *pesticides*;
… *country-specific* data of *freight and passenger transport* by rail of several European countries;
… data of new/additional means of transport of a person (*two-wheels, e-mobility, city cars*);
… data of new means of communication (i.e. *videoconference*);
… *additional electricity mixes* for Switzerland (representing the certified electricity production and the consumer mix, including and excluding the share of certified electricity);
… and, data of *coal tailings* in order to complete the tailings issue (sulphidic and uranium tailings model have been added in the framework of the update to version v2.1).
The list on the following page gives an overview of all new datasets.

(i) PESTICIDES		
aclonifen, at regional storage	RER	kg
captan, at regional storage	RER	kg
chlorothalonil, at regional storage	RER	kg
chlorotoluron, at regional storage	RER	kg
diazine-compounds, at regional storehouse	RER	kg
diazole-compounds, at regional storehouse	RER	kg
dimethenamide, at regional storage	RER	kg
folpet, at regional storage	RER	kg
fosetyl-Al, at regional storage	RER	kg
fungicides, at regional storehouse	RER	kg
growth regluators, at regional storehouse	RER	kg
herbicides, at regional storehouse	RER	kg
insecticides, at regional storehouse	RER	kg
isoproturon, at regional storage	RER	kg
mancozeb, at regional storage	RER	kg
mecoprop, at regional storage	RER	kg
metaldehyde, at regional storage	RER	kg
metamitron, at regional storage	RER	kg
napropamide, at regional storage	RER	kg
orbencarb, at regional storage	RER	kg
pendimethalin, at regional storage	RER	kg
prosulfocarb, at regional storage	RER	kg
pyridine-compounds, at regional storehouse	RER	kg
(ii) CHEMICALS / PLASTICS		
concentrated lithium brine (6.7 % Li), at plant	GLO	kg
ethylene carbonate, at plant	CN	kg
lithium fluoride, at plant	CN	kg
lithium hexafluorophosphate, at plant	CN	kg
Manganese dioxide oxidation	RER	kg
manganese oxide (Mn2O3), at plant	CN	kg
methylamine, at plant	RER	kg
phosphorous pentachloride, at plant	CN	kg
polystyrene foam slab, 100% recycled, at plant	CH	kg
polystyrene foam slab, 45% recycled, at plant	CH	kg
polystyrene scrap, old, at plant	CH	kg
potassium carbonate from manganese dioxide oxidation, at plant	RER	kg
potassium permanganate, at plant	RER	kg
(iii) ENERGY		
electricity, certified eletricity	CH	kWh
electricity, consumer mix	CH	kWh
electricity, high voltage, certified electricity. at grid	CH	kWh
electricity, high voltage, consumer mix, at grid	CH	kWh
electricity, low voltage, certified electricity, at grid	CH	kWh
electricity, low voltage, consumer mix, at grid	CH	kWh
electricity, medium voltage, certified electricity, at grid	CH	kWh
electricity, medium voltage, consumer mix, at grid	CH	kWh
electricity, wood, at distillery	SE	kg
ethanol, 95% in H2O, from wood, at distillery	SE	kg
ethanol, 99.7% in H2O, from Swedish wood, at service station	CH	kg
ethanol, 99.7% in H2O, from wood, at distillation	SE	kg
liquefied petroleum gas, at service station	CH	kg
petrol, 85% vol. ethanol, from Swedish wood, at service station	CH	kg
wood, in distillery	SE	kg
(iv) ELECTRONICS		
Anode, lithium-ion battery, graphite, at plant	CN	kg
Cathode, lithium-ion battery, lithium manganese oxide, at plant	CN	kg
chassis, network main devices	RER	kg
graphite, battery grade, at plant	CN	kg
network access devices, internet, at user	CH	unit
router, IP network, at server	CH	unit
separator, lithium-ion battery, at plant	CN	kg
single cell, lithium-ion battery, lithium manganese oxide/graphite, at pla	CN	kg
use, computer, laptop, videoconference	CH	h
use, computer, laptop, videoconference, certified electricity mix	CH	h
use, computer, laptop, work at home	CH	h
use, computer, laptop, work at home, certified electricity mix	CH	h
use, IP network, videoconference	CH	h
use, IP network, work at home	CH	h
use, network access devices	CH	h
use, network access devices, certified electricity	CH	h
videoconference, laptop, participant	CH	h
videoconference, laptop, participant, certified electricity mix	CH	h
work at home, corporate access	CH	h
work at home, corporate access, certified electricity mix	CH	h

(v) MOBILITY		
bicycle, at regional storage	RER	unit
disposal, bicycle	CH	unit
disposal, electric bicycle	CH	unit
disposal, electric scooter	CH	unit
disposal, electric vehicle, LiMn2O4	RER	unit
disposal, passenger car, diesel EURO5, city car	RER	unit
disposal, passenger car, electric, LiMn2O4, city car	RER	unit
disposal, scooter	CH	unit
electric bicycle, at regional storage	RER	unit
electric motor, electric vehicle, at plant	RER	kg
electric scooter, at regional storage	RER	unit
maintenance, bicycle	CH	unit
maintenance, electric bicycle	CH	unit
maintenance, electric scooter	CH	unit
maintenance, electric vehicle, LiMn2O4	RER	unit
maintenance, passenger car, diesel, EURO5, city car	RER	unit
maintenance, passenger car, electric, LiMn2O4, city car	RER	unit
maintenance, scooter	CH	unit
operation, average train	AT	pkm
operation, average train	IT	pkm
operation, average train	FR	pkm
operation, average train	DE	pkm
operation, average train	BE	pkm
operation, average train, SBB mix	CH	pkm
operation, electric bicycle	CH	km
operation, electric bicycle, certified electricity	CH	km
operation, electric scooter	CH	km
operation, electric scooter, certified electricity	CH	km
operation, freight train	AT	tkm
operation, freight train	IT	tkm
operation, freight train	FR	tkm
operation, freight train	DE	tkm
operation, freight train	BE	tkm
operation, freight train, diesel, with particle filter	CH	tkm
operation, high speed train	FR	pkm
operation, high speed train	IT	pkm
operation, metropolitan train, SBB mix	CH	pkm
operation, passenger car, diesel, EURO5, city car	CH	km
operation, passenger car, electric, LiMn2O4	CH	km
operation, passenger car, electric, LiMn2O4, certified electricity	CH	km
operation, passenger car, electric, LiMn2O4, city car	CH	km
operation, passenger car, electric, LiMn2O4, city car, certified electricity	CH	km
operation, scooter	CH	km
passenger car, diesel, EURO5, city car, at plant	RER	unit
passenger car, electric, LiMn2O4, at plant	RER	unit
passenger car, electric, LiMn2O4, city car, at plant	RER	unit
scooter, ICE, at regional storage	RER	unit
transport, average train	AT	pkm
transport, average train	IT	pkm
transport, average train	FR	pkm
transport, average train	DE	pkm
transport, average train	BE	pkm
transport, average train, SBB mix	CH	pkm
transport, bicycle	CH	pkm
transport, electric bicycle	CH	pkm
transport, electric bicycle, certified electricity	CH	pkm
transport, electric scooter	CH	pkm
transport, electric scooter, certified electricity	CH	pkm
transport, freight, rail	AT	tkm
transport, freight, rail	IT	tkm
transport, freight, rail	FR	tkm
transport, freight, rail	DE	tkm
transport, freight, rail	BE	tkm
transport, freight, rail, diesel, with particle filter	CH	tkm
transport, high speed train	FR	pkm
transport, high speed train	IT	pkm
transport, metropolitan train, SBB mix	CH	pkm
transport, passenger car, diesel, EURO5, city car	CH	pkm
transport, passenger car, electric, LiMn2O4	CH	pkm
transport, passenger car, electric, LiMn2O4, certified electricity	CH	pkm
transport, passenger car, electric, LiMn2O4, city car	CH	pkm
transport, passenger car, electric, LiMn2O4, city car, certified electricity	CH	pkm
transport, scooter	CH	pkm
(vi) WASTE TREATMENT		
disposal, spoil from coal mining, in surface landfill	GLO	kg
disposal, spoil from lignite mining, in surface landfill	GLO	kg
disposal, tailings from hard coal milling, in impoundment	GLO	kg

2. NEW IMPACT ASSESSMENT METHODS:

Two new LCIA methods (USEtox, ReCiPe) have been included; for the latter one three different weighting schemes on both levels (midpoints, endpoints) have been included. More information about these methods can be found (soon) in the updated ecoinvent report No.3, the implementation report of the LCIA methods.

3. CHANGES IN IMPACT ASSESSMENT METHODS

A harmonisation step has been undertaken concerning the issue of the long-term emissions (resp. their valuation), as in version 2.1, the issue has been implemented in two different ways. In version 2.2, for the following methods, two different implementations can be found in the database – one including factors for long-term emissions; and a second one without these factors:

- CML method,
- Eco-Indicator'99 (only the Hierarchist and the Egalitarian perspective),
- EDIP (version 97 and 2003),
- ReCiPe (Midpoints and Endpoints – each again only for Hierarchist and Egalitarian perspective),
- USEtox

More information about the use of the varous implementations of these LCIA methods can be found (soon) in the updated ecoinvent report No.3, the implementation report of the LCIA methods.

4. FURTHER CHANGES

All the errors discovered & identified since the publication of the last version of the database (i.e. data v2.1) have been corrected for this new version of the database.
In addition, double entries of several chemical products (helium, 1-butanol, acrylonitrile) have been eliminated and several punctual updates (insulation materials, biogas mix Switzerland) have been integrated into the database.

A more detailled overview of all these corrections and changes – as well as a complete documentation of the new datasets – can be found (soon) in the latest update of report No.16, the report about the changes between v2.01, v2.1 and v2.2.